

OMAR ABDOU'S CV

BASIC INFORMATION

- **Name:** Omar Abdou
- **Title:** Senior Software Engineer | Distributed Systems & Backend Architecture | AI Product Builder
- **Email:** omar.abdo.jobs@gmail.com **Phone number:** +491629082097
- **Website** omar-abdo.com **LinkedIn:** [/omar-abdo](https://www.linkedin.com/in/omar-abdo) **Github:** [/OmarAbdo](https://github.com/OmarAbdo)

PROFESSIONAL SUMMARY

With **over 8 years of experience** designing scalable solutions in fintech, SaaS, and AI, I am a full-stack software developer (leaning backend) with a particular passion for AI and **distributed systems**. Applying concepts such as distributed consensus, consistency models, replication, and fault-tolerant system design from scratch in **my own distributed key-value database, Crystal DB**. As co-founder and lead engineer at LifeOS Labs, I architected Quik AI & Tafkeer AI — AI aggregator platforms integrating 44+ LLMs with vector and graph RAG memory, alongside authentication, payment, and compliance systems — work I built while completing a Master's in Data Science to deepen my ML foundations. A proven ability to work with limitations and legacy systems and apply incremental migration/change (e.g., **from monolithic to microservices**) with opinionated architectural choices across different levels of the product, as well as leading impactful projects end-to-end, such as a gym sales system that became one of the three most profitable among the company's 40 international locations. An Expert in modern tech stacks such as Go, NodeJs, Python, React, and microservices, AI, and event-driven architecture, combining a unique set of strong technical and leadership skills, having managed **my own software agency, Code Netro**, that **handled over 12 projects in five different countries**.

SKILLS

- **Programming Languages:** C, C++, Java, Go, Python, JavaScript, TypeScript, PHP
- **Distributed Systems:** State Machine Replication, Distributed Logs, Apache Kafka, Consensus Algorithms, Logical & Physical Clocks, Coordination Services, CAP Theorem, Consistency Models, Asynchronous Communication.
- **DevOps, Cloud & Infrastructure:** Docker, Kubernetes, Terraform, AWS, GitHub Actions
- **Backend & Systems Architecture:** REST/Restful API, gRPC, Event-Driven Architecture, Event Streaming & Messaging, Microservices, Domain-Driven Design (DDD), Go (Gin), Python (FastAPI, Django, Flask), Node.js (Express, NestJS), PHP
- **Databases:** Storage Engine Internals (B-Trees, LSM-Trees), ACID & Isolation Levels, Schema Encoding & Evolution, Replication & Sharding, PostgreSQL, Redis, MongoDB (NoSQL).
- **Frontend Development:** React, Next.js, Vue, Nuxt.js, and Vite

EXPERIENCE

Co-Founder & Lead Engineer

LifeOS Labs, Berlin, Jan 2025 – Present

- **Crystal DB:** A **distributed consensus-based** key-value database built from scratch in Go. Implements the **RAFT** consensus algorithm end-to-end. Including **leader election**, **log replication**, with a crash-safe **write-ahead log**, snapshot-based **log compaction** and a single-writer control loop driving per-peer replication goroutines. State machine is pluggable via a storage interface designed for **LSM-tree persistent storage**. Validated with unit tests and a real multi-node cluster integration suite **covering leader failure and data survival**.

- **Quik**: LifeOS's first AI aggregator and super-app, enabling users to chat, search, and research across **44+ LLMs** with full **vector- and graph-RAG memory**. As lead engineer I directed the complete lifecycle — from Figma design to deployment and DevOps. Built with Go, PostgreSQL, a **microservices backend**, and React.
- **Tafkeer AI**: AI aggregator with an autonomous routing layer selecting across 59 LLMs and specialized assistants — **the first of its kind in the region**. Extended the Quik platform architecture and adapted it for the Saudi market.

Independent Software R&D

Open-source, Remote / Berlin, Oct 2023 – Dec 2024

- **Netro**: An object-oriented **Python simulation** framework benchmarking hybrid Last-Mile Delivery (LMD) systems — a mix of autonomous delivery robots and trucks — against traditional methods. Built with **NumPy, Pandas, and Domain-Driven Design**.
- **Luqman**: An **LSTM recurrent-neural-network** trading assistant for automated market analysis and time-series forecasting, applying sequential deep-learning models to predictive decision-making. Built in **Python**.

Frontend developer

Kurnblick, Paderborn, part time, Oct 2021 – September 2023

- **Papenburg's Events management system**: A Rest API based system that streamlines the processes of event creation, handling customer data, ticketing, and digital payments. Allowing the organization to *manage more events with less* resources. Developed with **Vue.js, Nuxt.js, and Typescript**.

Backend developer

Simon Kucher, Bonn, Full time, Aug 2019 - Apr 2022

- **MacFit sales system**: An End-to-end sales system with a sophisticated recommendation engine and sales funnel tracking features of each custom across the multiple gym branches. I took the **initiative to step up as the project manager**. MacFit ended up *being the 3rd most profitable project throughout the company's international 40 offices* of the year. Developed with **Node.js, Vue.js, PostgreSQL, and Docker**
- **Precision Pricing**: A forecasting and optimization engine and an ETL designed for the Royal Bank of Canada. Developed with **Python**

Founder & Full stack Software developer

Code Netro, Online, Jan 2016 - July 2019

- **CodeNetro**: Started my own software development agency to keep up with the increased demand over my successful software development services and quality. During Code Netro more than **12 projects were delivered over 5+ countries**. Beneath are some honorable mentions. Mainly developer with **React, PHP, Laravel, NodeJs, VueJs, and MySQL**
- **Ridee** — Developed a Mobile app to arrange cycling communities & events in Berlin organized by Ridee Gmbh. Developed with **Dart & Flutter**
- **Eonity's Pitch Master**: A music scheduler & trading platform based on Spotify's API and owned by the music label Eonity. Developed with **Javascript, Typescript, React, NodeJs, NestJs, and Mongo DB**
- **SunnyWorld**: Developed a commercial and an educational platform for the entrepreneurial company Sunny world. Transforming the company from a 2-people project into a **competing company with multiple offices** in the renewable energy sector. Developed with **PHP and MySQL**

EDUCATION

Masters of Data Science

International Business Administration Faculty, European University Viadrina Frankfurt (Oder) April 2022 – August 2025

- Data Science, Machine Learning, Data Visualization, Predictive Analytics, Statistical Modeling, Supply Chain, Forecasting, and SQL Databases

Bachelor of Business Administration

Business Faculty, Tanta University July 2016 – September 2021

- Business Administration, Organizational Behavior, Applied statistics, Project Management, Entrepreneurship, Business Digitalization, Database management

LANGUAGES

- English
- Arabic (Egypt)
- German

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Then feel free to reach out to my [LinkedIn](#) and I'd be happy to answer your questions!